THE DUAL ISOGENY

In this lecture, we associate to an isogeny an isogeny in the opposite direction. An isogeny is usually not invertible, but its dual is the next best thing: composing the two maps gives multiplication by the degree. We will first construct this reverse map, and afterwards interpret it in terms of divisor classes. The resulting identities will explain the quadratic behaviour of degree.

Throughout, K is a perfect field and $k = \bar{K}$. Elliptic curves are defined over K, while points, kernels, and divisors are taken over k unless otherwise specified.

1. What should a dual isogeny do?

Let $\phi : E_1 \to E_2$ be a nonzero isogeny, and put $m = \deg \phi$. Given $Q \in E_2$, choose $P \in E_1$ with $\phi(P) = Q$. The proposed reverse map is

$$Q \longmapsto [m]P.$$

Why should this be independent of P? Any other lift is $P + T$ for $T \in \ker \phi$, and $\#\ker \phi = \deg_s \phi$ divides m. Thus $[m]T = O_1$. This gives a well-defined map of groups on geometric points. However, *a well-defined map on points is not yet a morphism of curves.* Establishing this last assertion is the main task.

2. Existence and uniqueness

Theorem 2.1 (The dual isogeny)**.** *Let $\phi : E_1 \to E_2$ be a nonzero isogeny of degree m. There is a unique isogeny $\widehat{\phi} : E_2 \to E_1$ such that*

$$\widehat{\phi} \circ \phi = [m]_{E_1}. \tag{2.1}$$

It also satisfies $\phi \circ \widehat{\phi} = [m]_{E_2}$. If ϕ is defined over K, then so is $\widehat{\phi}$.

Proof. We first work over k.

The separable case. If ϕ is separable, then $\Phi = \ker \phi$ has order m. Thus $[m]$ kills Φ. Lemma **??** gives a morphism $\widehat{\phi}$ satisfying (2.1). It preserves the distinguished points and is nonconstant, since $[m]$ is nonconstant. This proves existence in the separable case, even if the characteristic divides m.

The Frobenius step. Suppose now that $\operatorname{char} K = p > 0$. For $q = p^r$, write $F_q : E \to E^{(q)}$ for relative Frobenius. We claim that there is a homomorphism

$$V_q : E^{(q)} \longrightarrow E, \qquad V_q \circ F_q = [q].$$

Indeed, $[p]$ is nonconstant and $[p]^*\omega = p\omega = 0$, so $[p]$ is inseparable. Write $\deg_i [p] = p^a$ with $a \geq 1$. Since $[q]$ is the r-fold composite of $[p]$, its inseparable degree is p^{ar}. Frobenius factorisation therefore gives

$$[q] = \eta \circ F_{p^{ar}}.$$

Since $ar \geq r$, the Frobenius map on the right factors as

$$E \xrightarrow{F_q} E^{(q)} \xrightarrow{F_{p^{ar-r}}} E^{(p^{ar})}.$$

Composing the second arrow with η gives V_q. It is nonconstant and sends $O^{(q)}$ to O, and hence is a homomorphism. When $r = 0$, take $F_1 = V_1 = [1]$. Notice that we have not used $\deg[q] = q^2$.

The general case. Factor ϕ as

$$E_1 \xrightarrow{F_q} E_1^{(q)} \xrightarrow{\lambda} E_2, \qquad q = \deg_i \phi,$$

where λ is separable of degree $s = m/q$. The separable case supplies $\widehat{\lambda} : E_2 \to E_1^{(q)}$ with $\widehat{\lambda} \circ \lambda = [s]$. Set

$$\widehat{\phi} = V_q \circ \widehat{\lambda}.$$

Because homomorphisms commute with multiplication by integers,

$$\widehat{\phi} \circ \phi = V_q \circ [s] \circ F_q = [s] \circ V_q \circ F_q = [s] \circ [q] = [m].$$

Uniqueness and the other composition. If $\psi \circ \phi = [m]$, surjectivity of ϕ gives $\psi = \widehat{\phi}$. Similarly,

$$(\phi \circ \widehat{\phi}) \circ \phi = \phi \circ [m] = [m] \circ \phi,$$

so cancellation of the surjective map ϕ gives $\phi \circ \widehat{\phi} = [m]$.

The ground field. If ϕ is defined over K, conjugating (2.1) by $\sigma \in G_K$ gives $\widehat{\phi}^{\sigma} \circ \phi = [m]$. Uniqueness implies $\widehat{\phi}^{\sigma} = \widehat{\phi}$. Since K is perfect, the Galois criterion for a rational map to be defined over K finishes the proof. □

Definition 2.2. The isogeny in Theorem 2.1 is called the *dual isogeny* of ϕ. We extend the notation to the zero homomorphism by setting $\widehat{[0]} = [0]$.

Remark 2.3. A kernel of geometric points determines a separable quotient, but not an arbitrary inseparable isogeny. For example, F_q has only O in its geometric kernel although its degree is q. This is why the general existence proof needs a Frobenius step, rather than just an argument with finite groups of points.

3. Why the construction is dual: pullback of divisors

Recall the group isomorphism

$$\kappa_E : E(k) \xrightarrow{\sim} \mathrm{Pic}^0(E), \qquad P \longmapsto \mathrm{cl}\big((P) - (O)\big).$$

Here $\mathrm{Pic}^0(E)$ is simply the group of degree-zero divisor classes on the curve over k; no algebraic-variety structure on this group is being used.

Proposition 3.1. *For every nonzero isogeny $\phi : E_1 \to E_2$,*

$$\kappa_{E_1} \circ \widehat{\phi} = \phi^* \circ \kappa_{E_2}. \tag{3.1}$$

Thus the dual isogeny corresponds to pullback on degree-zero divisor classes.

Proof. Put $q = \deg_i \phi$, $\Phi = \ker \phi$, and $s = \#\Phi = \deg_s \phi$, so $m = qs$. Choose P above $Q \in E_2(k)$. The fibres over Q and O_2 are $P + \Phi$ and Φ, respectively. Every point has ramification index q, so

$$\phi^*\big((Q) - (O_2)\big) = q \sum_{T \in \Phi} \big((P+T) - (T)\big).$$

Each summand has divisor class $\kappa_{E_1}(P)$. The class of the entire expression is therefore

$$qs\,\kappa_{E_1}(P) = \kappa_{E_1}([m]P) = \kappa_{E_1}(\widehat{\phi}(Q)).$$

This proves (3.1). □

We have now justified both parts of the construction: the reverse map is a morphism, and its action on points is precisely the action of pullback on divisor classes. These are distinct assertions.

Corollary 3.2. *Let $R, S \in E_2(k)$ and let ϕ be nonzero. Then*

$$\mathrm{cl}\big(\phi^*(R) - \phi^*(S)\big) = \kappa_{E_1}\big(\widehat{\phi}(R - S)\big).$$

Moreover, as sets of geometric points,

$$\ker \widehat{\phi} = \phi(E_1[m]), \qquad m = \deg \phi.$$

Proof. The first assertion follows by subtracting (3.1) for R and S. For the second, write $Q = \phi(P)$. Then $\widehat{\phi}(Q) = O_1$ if and only if $[m]P = O_1$. □

4. The algebra of duality

We begin with the properties that follow directly from uniqueness.

Proposition 4.1. *Let $\phi : E_1 \to E_2$ and $\lambda : E_2 \to E_3$ be homomorphisms. Then*

$$\widehat{\lambda \circ \phi} = \widehat{\phi} \circ \widehat{\lambda}.$$

If $\alpha : E_1 \to E_2$ is an isomorphism preserving the distinguished points, then $\widehat{\alpha} = \alpha^{-1}$. In particular, $\widehat{[-1]} = [-1]$ and $\widehat{-\phi} = -\widehat{\phi}$.

Proof. The proof is fairly elementary as we discussed in class. □

Theorem 4.2 (Additivity)**.** *For $\phi, \psi \in \mathrm{Hom}(E_1, E_2)$,*

$$\widehat{\phi + \psi} = \widehat{\phi} + \widehat{\psi}.$$

This is the one property that does not follow merely by cancelling an isogeny in the defining identities. Appendix A gives an elementary proof: compare the divisors of

$$x(\phi(P) + A) - x(\psi(P) + B)$$

and of the same function after translating both A and B by a point T. The comparison gives

$$[2] \circ \widehat{\phi + \psi} = [2] \circ (\widehat{\phi} + \widehat{\psi}).$$

A nonzero homomorphism of elliptic curves is surjective, so [2] can be cancelled. The appendix supplies the local multiplicity calculation and explains why this works in characteristic two as well. No theorem about Picard varieties or the theorem of the square is needed.

Theorem 4.3. *For every integer n and every homomorphism $\phi : E_1 \to E_2$,*

$$\widehat{[n]} = [n], \qquad \deg[n] = n^2, \qquad \deg \widehat{\phi} = \deg \phi, \qquad \widehat{\widehat{\phi}} = \phi.$$

Proof. Since $\widehat{[1]} = [1]$, additivity and the formula for negatives imply $\widehat{[n]} = [n]$ for all n. For $n \neq 0$, the defining identity then gives

$$[\deg[n]] = \widehat{[n]} \circ [n] = [n^2].$$

The injectivity of $\mathbb{Z} \to \mathrm{End}(E)$ implies $\deg[n] = n^2$. The assertion for $n = 0$ is our convention.

If $\phi \neq [0]$ has degree m, multiplicativity of degrees gives

$$m \deg \widehat{\phi} = \deg(\widehat{\phi} \circ \phi) = \deg[m] = m^2.$$

Thus $\deg \widehat{\phi} = m$. Since $\phi \circ \widehat{\phi} = [m]$, uniqueness of the dual of $\widehat{\phi}$ gives $\widehat{\widehat{\phi}} = \phi$. Again, the zero case is immediate. □

Remark 4.4. The order of the proof matters. We constructed the dual without using $\deg[n] = n^2$; proved its divisor interpretation; proved additivity independently; and only then deduced the degree formula. In particular, none of these arguments uses Tate modules or the Weil pairing.

5. Degree is a positive definite quadratic form

For $\phi, \psi \in \mathrm{Hom}(E_1, E_2)$, define the integer

$$B(\phi, \psi) = \deg(\phi + \psi) - \deg \phi - \deg \psi.$$

Theorem 5.1. *The pairing B is symmetric and additive in each variable. Moreover,*

$$[B(\phi, \psi)] = \widehat{\phi} \circ \psi + \widehat{\psi} \circ \phi \qquad \text{in } \mathrm{End}(E_1). \tag{5.1}$$

Consequently the function $\deg : \mathrm{Hom}(E_1, E_2) \to \mathbb{Z}$ is a positive definite quadratic form: it is unchanged by negation, its associated pairing is bilinear, and $\deg \phi > 0$ for $\phi \neq [0]$.

Proof. Expand using additivity of the dual:

$$\begin{aligned}[\deg(\phi+\psi)] &= (\widehat{\phi}+\widehat{\psi})\circ(\phi+\psi)\\ &= [\deg\phi]+[\deg\psi]+\widehat{\phi}\circ\psi+\widehat{\psi}\circ\phi.\end{aligned}$$

This proves (5.1). Its right-hand side is additive in both variables. Since $\mathbb{Z}\to\mathrm{End}(E_1)$ is injective, B itself is bilinear. Symmetry is immediate. Finally, $\deg(-\phi)=\deg\phi$ because negation has degree one, and every nonconstant morphism of curves has positive degree. □

Corollary 5.2 (Parallelogram identity)**.** *For every $\phi,\psi\in\mathrm{Hom}(E_1,E_2)$ and every $n\in\mathbb{Z}$,*

$$\deg(\phi+\psi)+\deg(\phi-\psi)=2\deg\phi+2\deg\psi,\qquad \deg(n\phi)=n^2\deg\phi.$$

Proof. Use $B(\phi,-\psi)=-B(\phi,\psi)$ for the first identity. For the second, $n\phi=[n]\circ\phi$, so degrees multiply. □

Thus degree behaves like the square of a length, rather than like a linear function. Notice the distinction between $n\phi$, which is a sum of homomorphisms, and an n-fold composition; the latter is only defined when the sources and targets match appropriately.

6. Differentials and the dual of Frobenius

Assume $\phi\neq[0]$, and choose invariant differentials $\omega_1,\omega_2\neq 0$. Write

$$\phi^*\omega_2=a_\phi\omega_1,\qquad \widehat{\phi}^*\omega_1=a_{\widehat{\phi}}\omega_2.$$

Proposition 6.1. *If $m=\deg\phi$, then $a_\phi a_{\widehat{\phi}}=m$ as an equality in k.*

Proof. Pull back ω_1 along $\widehat{\phi}\circ\phi=[m]$:

$$m\omega_1=[m]^*\omega_1=\phi^*(\widehat{\phi}^*\omega_1)=a_{\widehat{\phi}}a_\phi\omega_1.$$

□

If the characteristic does not divide m, both scalars are nonzero, so both isogenies are separable. If the characteristic divides m, at least one scalar is zero, so at least one of the two isogenies is inseparable. This is an equality in k, not an equality of integer degrees.

Definition 6.2. In characteristic $p>0$, the dual of relative Frobenius

$$F_{E/K}:E\longrightarrow E^{(p)}$$

is called the *Verschiebung*, and is denoted $V_{E/K}:E^{(p)}\to E$.

The construction in Theorem 2.1 is precisely this map. We now also know its degree:

$$\deg F_{E/K}=\deg V_{E/K}=p,\qquad V_{E/K}\circ F_{E/K}=[p]_E,\qquad F_{E/K}\circ V_{E/K}=[p]_{E^{(p)}}.$$

Frobenius is purely inseparable, but its dual need not be. Indeed, $\deg_s[p]=\deg_s V_{E/K}$, so the number of geometric p-torsion points is either p or 1, according as $V_{E/K}$ is separable or inseparable. We will return to this distinction when studying torsion in characteristic p.

Appendix A. An elementary proof of additivity

The point of this appendix is to justify Theorem 4.2 using rational functions and their divisors.

Fix a Weierstrass x-coordinate on E_2. We know that its pole divisor is $2(O_2)$ and that

$$x(U)=x(W)\quad\Longleftrightarrow\quad W=U\text{ or }W=-U$$

for affine points. The map $x:E_2\to\mathbb{P}^1$ is separable of degree two: its nontrivial automorphism is negation. At a finite point R not killed by 2, the fibre consists of the two distinct points R and $-R$, so the degree formula for fibres shows that x is unramified there. These statements also hold in characteristic two. The set $E_2[2]$ is finite by Appendix **??**; no formula for its order is needed.

For a nonzero homomorphism $h:E_1\to E_2$, use the notation

$$D_{h,R}=h^*(R).$$

If $h = [0]$ and $R \neq O_2$, put $D_{h,R} = 0$, the divisor of the empty fibre. We will never use $D_{[0],O_2}$.

Lemma A.1 (The x-coordinate divisor identity)**.** *Let $\phi, \psi : E_1 \to E_2$ be nonzero homomorphisms. For suitably chosen $A, B \in E_2(k)$, the rational function*

$$H_{A,B}(P) = x(\phi(P) + A) - x(\psi(P) + B)$$

is nonzero and has divisor

$$\operatorname{div}(H_{A,B}) = D_{\phi-\psi,B-A} + D_{\phi+\psi,-A-B} - 2D_{\phi,-A} - 2D_{\psi,-B}. \tag{A.1}$$

The points A, B may be chosen so that the same formula also holds with A, B replaced by $A + T, B + T$ for all but finitely many $T \in E_2(k)$.

Proof. Write $f = \phi + A$ and $g = \psi + B$. Choose A, B so that f and g never simultaneously equal the same point of $E_2[2]$, including O_2, and so that neither $f - g$ nor $f + g$ is identically O_2. Such a choice exists. After fixing A, there are only finitely many points of E_1 at which $\phi(P) + A$ belongs to the finite set $E_2[2]$, and each forbids only one value of B. The two possible identities forbid at most two further values.

The poles of $H_{A,B}$ occur where $f = O_2$ or $g = O_2$. These loci do not overlap, so their orders give exactly $2D_{\phi,-A} + 2D_{\psi,-B}$. At a zero, $f = g$ or $f = -g$, and the common x-coordinate is not a branch value at either of these points. The two possibilities do not overlap.

We also need the multiplicities. Near a pair (R, R) with $R \notin E_2[2]$, the function $x(U) - x(W)$ is a unit multiple of a uniformizer at O_2 evaluated at $U - W$. One can see this in local coordinates: put $Z = U - W$, and expand $x(W + Z) - x(W)$ in a parameter at $Z = O_2$. The constant term is zero and the linear coefficient is nonzero, since translation is an isomorphism and x is a parameter at R. Thus the quotient by that parameter is a unit. The same calculation near $(R, -R)$ uses $U + W$, since $x(-W) = x(W)$. Pulling these local identities back to E_1 gives precisely the ramification multiplicities in the first two terms of (A.1). This argument allows inseparable $\phi \pm \psi$: the higher orders arise on pulling back the parameter.

Finally, simultaneous values in $E_2[2]$ after a common translation can occur only at points where

$$(\phi - \psi)(P) = B - A.$$

If $\phi - \psi \neq [0]$, this is a finite set; if it is zero, our choice $A \neq B$ makes the set empty. Hence only finitely many T create such a value. If $\phi + \psi = [0]$, also exclude the finitely many solutions of $[2]T = -A - B$. This proves the last assertion. □

Proof of Theorem 4.2. If one of ϕ, ψ is zero, there is nothing to prove. Otherwise choose A, B as in Lemma A.1. Subtract (A.1) from the corresponding identity for $A + T, B + T$. The $\phi - \psi$ terms cancel. Since the difference is principal, we obtain

$$\begin{aligned} &D_{\phi+\psi,-A-B-2T} - D_{\phi+\psi,-A-B} \\ &\qquad \sim 2\big(D_{\phi,-A-T} - D_{\phi,-A}\big) + 2\big(D_{\psi,-B-T} - D_{\psi,-B}\big). \end{aligned}$$

For a nonzero h, Corollary 3.2 says that

$$\operatorname{cl}(D_{h,R} - D_{h,S}) = \kappa_{E_1}(\widehat{h}(R - S)).$$

It is also true for $h = [0]$ whenever the two displayed fibres are defined, since then both sides are zero. Therefore the divisor relation gives

$$\widehat{\phi + \psi}(-2T) = [2]\widehat{\phi}(-T) + [2]\widehat{\psi}(-T).$$

Each dual is already a homomorphism. Cancelling the minus signs gives

$$[2]\big(\widehat{\phi + \psi}(T) - \widehat{\phi}(T) - \widehat{\psi}(T)\big) = O_1$$

for all but finitely many T. Equality of morphisms on a dense open subset implies equality everywhere. Put $\delta = \widehat{\phi + \psi} - \widehat{\phi} - \widehat{\psi}$; we have $[2] \circ \delta = [0]$. If δ were nonzero, it would be surjective and would force $[2] = [0]$, contradicting Appendix **??**. Thus $\delta = [0]$. Note that this step does not divide a field element by 2. It cancels a nonzero homomorphism, so the proof is valid in characteristic two. □